

AI-Powered Response Authenticity Detection for Hiring

Project Description and Clarification

Symbol



Symbol Jr.

Damien Flutre

Ziming Fu

Monica Kim

Vsevolod (V) Kovalev

Khang Bui

Table of Contents

I. Introduction 3

 I. 1. Project Introduction.....3

 I. 2. Background and Related Work3

 I. 3. Project Overview4

 I. 3. Client and Stakeholder Identification and Preferences.....5

II. Glossary 6

III. References..... 6

I. Introduction

I.1. Project Introduction

As hiring shifts toward remote and asynchronous interviews, powerful AI tools make it easy for candidates to lean on generated text, read polished scripts, or even play back synthetic speech. This blurs the signal that reviewers rely on to judge real communication skills and in the moment reasoning, creating a trust gap for platforms like Symbal's AI-enabled process. Our motivation is to restore trust by giving reviewers evidence about authenticity rather than guesses.

Single channel detectors are not enough. Text only methods that look for model-like writing patterns can be brittle on short, lightly edited answers, and watermarking only helps when the model is under our control. On the audio side, deepfake and TTS detectors can miss unseen synthesis methods and noisy real-world recordings. Research on read versus spontaneous speech, however, shows prosodic and disfluency cues that help distinguish scripted delivery from genuine answers, suggesting value in fusing complementary signals rather than trusting any one test. Thus, it is increasingly important to create dedicated systems tailored to detect AI or script usage.

I.2. Background and Related Work

Asynchronous, AI-assisted interviews create a verification problem: a candidate can (a) paste or lightly edit model-generated text, (b) read a prepared script, or (c) play back synthetic text-to-speech (TTS). Prior tools that try to flag AI-written text have struggled with accuracy, especially on short answers and edited text. OpenAI, for example, retired its AI-text classifier in 2023 due to a low accuracy rate on real-world inputs [1]. The takeaway is simple: a single "AI vs. human" detector should not be the only decision tool in a high-stakes setting like hiring.

Most text-only detectors look for statistical patterns typical of machine output. GLTR, for instance, checks how "predictable" each word is to a language model - often called perplexity (how surprising a piece of text is to the model) and related rank signals [2]. Newer methods (DetectGPT, Fast-DetectGPT) test how a model's confidence changes when we make small edits - called perturbations to the text; if the score changes in a certain way, the text is more likely to be machine-written [3], [4]. These signals help, but they are fragile: simple paraphrasing can defeat them, and short answers (common in interviews) reduce reliability even more [1], [5].

Another line of work focuses on where the content came from its source. Watermarking nudges a model to choose certain tokens so its outputs carry a detectable signature [6]. This works under controlled conditions, but paraphrasing can weaken the signal, and it only helps if we control the model the candidate used, which we do not in open hiring contexts [5]. A practical alternative is retrieval-based matching: compare a candidate's answer to a large library of known model outputs and flag close matches, even after paraphrase. This suggests a design that combines several signals rather than relying on a single detector.

For audio, the closest mature area is deepfake/voice-spoof detection and speaker-verification anti-spoofing. The ASVspoof benchmarks (2019, 2021) defined attack types and metrics that drove progress in end-to-end models like RawNet2 and graph-based models like AASIST [7], [8], [9]. Results from later challenges (e.g., ADD 2023) show a key risk for hiring use: systems trained on a few TTS engines may miss unseen vocoders or real-world recording conditions. Good generalization and careful calibration are therefore important; deepfake detection should be regarded as one input signal, not the sole judge [10].

A complementary literature distinguishes read vs. spontaneous speech - exactly our “reading from a script vs. genuine answer” question. Studies in phonetics and prosody (the rhythm, pitch, and timing of speech) show some patterns: spontaneous speech has more disfluencies (um/uh, repeats, repairs), more varied timing (speaking-rate changes and pause placement), and different pitch patterns; read speech is smoother, more regular, and has fewer disfluencies [11], [12], [13], [14], [15]. Classifiers that use simple prosodic features - pitch range, inter-pausal units, articulation rate - can separate these styles above chance. This supports extracting prosody/disfluency features from interview audio and combining them with the text from ASR to score “scriptedness.”

The team will learn: core speech processing (voice activity detection, feature extraction, ASR alignment), anti-spoofing metrics (ASVspoof/ADD), implementation limits of text detectors (GLTR, DetectGPT/Fast-DetectGPT) under paraphrasing, and evaluation with ROC/DET, equal-error-rate, and cost-sensitive thresholds. We will learn about bias testing (especially for non-native speech/writing), and clear documentation of limitations. On the systems side, we will learn how to containerize a FastAPI/Flask microservice, and how to produce reproducible evaluation scripts so Symbal can validate the authenticity score on their data [2], [3], [4], [7], [10].

I.3. Project Overview

With the current implementation of Symbal’s AI-powered candidate hiring solution, there remains a question of candidates using scripts, pre-recorded responses, or AI in order to answer asynchronous interview questions as best as possible. In recent years, there has been an influx of candidates using sophisticated AI tooling to support their hiring process and interviews [16]. These tools are widely available and, while powerful, they create a growing sense of mistrust between candidates and hiring managers. Organizations want to ensure that they are hiring people for their authentic skills and communication abilities, not their ability to prompt a chatbot.

This problem raises a critical need: employers and hiring platforms require a reliable way to verify the authenticity of candidate responses. If this trust gap is not addressed, hiring outcomes may be distorted, and the credibility of AI-enabled recruiting platforms could be undermined. By introducing a system that can algorithmically assess whether answers are human-generated, scripted, or AI-assisted, Symbal can provide a higher level of confidence in its candidate screening process.

The goal of our development effort is to design and implement an AI-powered integrated API service that will detect whether spoken or written responses during interviews originated from a human, or a script/AI system. This service will work alongside and support the existing asynchronous AI candidate hiring process which Symbal has developed, and be able to restore confidence in digital interviews, reduce unfair advantages from AI misuse, and ultimately improve hiring outcomes.

The system will be designed as an API, which will allow for easy integration and deployment into Symbal’s existing framework. This API will act as a microservice and provide REST API calls which will accept either audio or text input from various applications, process the data if necessary, and then pass it through trained AI models. Then, the AI model will analyze and return a confidence score. This approach allows the API to exist independently from either side, meaning it is immune to changes in the AI models, and the services which call the API’s REST functions.

The API will support textual analysis through existing real-time AI-texting services, as well as post-call analysis pipelines to bolster the candidate filtering and analysis process. Our tooling will be able to provide actionable and measurable confidence scores based on the authenticity of candidate responses. For text, the API will be queried using standard strings, which will then be analyzed and return a confidence score, along with label(s) containing information about the

confidence score. For audio, the API will be queried using .mp3 or .wav audio blobs, which will be processed as necessary, and then acted on the same as text inputs.

A key strength of this design is its multi-modal capability. The system will support both text and audio detection, making it applicable across different stages of the hiring process. Text detection will be prioritized first, using methods that draw on existing AI-detection tools that examine patterns in language such as fluency, predictability, and repetition. Audio detection will follow, examining how responses are spoken by looking at natural variation in tone and pacing. Together, these two modes provide broad coverage for both written and spoken interviews.

The service is also being built to be scalable and flexible. Because it is structured as a microservice with REST API endpoints, it can easily integrate with other applications beyond Symbol's platform, such as third-party recruitment systems or post-call analysis pipelines. Containerization with Docker will further support portability and deployment flexibility, ensuring that the service can be run in cloud environments or on-premises as needed.

From a technical standpoint, the system combines adapted AI-detection models with feature-based classifiers. For text inputs, it will build on recent detection approaches like GPTZero and DetectGPT. For audio, it will rely on classifiers trained to recognize the difference between real human speech and computer-generated voices. Although these methods are technically complex, the intended outcome is straightforward: to provide hiring managers with clear and easy-to-understand indicators about whether a response is authentic.

We will build a multi-modal authenticity score that fuses: (1) text signals (perplexity/burstiness, small-edit tests, repetition patterns, and retrieval checks), (2) audio signals (prosody/disfluency cues for read vs. spontaneous; a deepfake/TTS probability), and (3) context signals (answer length, retakes, and response latency). We will also address fairness: some text detectors over-flag non-native writing [17]. To mitigate this, we will calibrate thresholds to keep false positives low, so Symbol can set policy-appropriate rules.

The aim is to deliver a production-ready service that can reliably operate at scale. In practice, this means supporting post-call analysis by flagging responses that might be AI-generated, as well as screening text replies in real time before they move further in the hiring process. Over time, the service will also provide useful data about how often AI is being used in recruiting, which can help improve the platform and inform broader research.

Ultimately, this project is about addressing a growing challenge in modern hiring: ensuring that while AI brings speed and efficiency, human evaluation remains fair and trustworthy. Success will be measured not just by technical accuracy, but by whether the service helps hiring managers make better decisions and builds confidence in the overall process.

I.4. Client and Stakeholder Identification and Preferences

Our primary client is Symbol, represented by Pranay Manocha, the company's founder and CTO. Symbol operates in the recruitment technology space, providing AI-powered candidate screening solutions, and requires our AI detection service to maintain the integrity of their interview platform. Pranay needs a reliable, scalable API that can seamlessly integrate with their existing infrastructure to identify when candidates use AI assistance, scripts, or human-originated during interviews.

Throughout the development process, the team will collaborate closely with Pranay to ensure the solution meets Symbol's technical requirements and expectations. Regular communication with Pranay will allow us to gather feedback, validate the approach, and adjust as needed. The end

product of this project will be a deployable API service that Pranay and his development team can integrate into their existing platform. We will focus on delivering a well-documented, production-ready solution that demonstrates best practices in AI detection while providing clear value to Symbal's recruitment technology offerings.

Recruitment professionals and hiring managers who use Symbal's platform depend on authentic candidate assessments to make informed hiring decisions. They need confidence that our detection system maintains high accuracy while minimizing false positives that could unfairly penalize legitimate candidates. Job candidates themselves are affected by our systems, as they need assurance that the detection operates fairly and transparently, without introducing bias against non-native speakers or penalizing natural speech patterns. By addressing these diverse needs and concerns, our AI detection service will support the integrity and fairness of the recruitment process.

II. Glossary

TTS (Text-To-Speech): Technology that enables text to be converted into speech sounds imitating the human voice.

GLTR (Giant Language model Test Room): See [2].

ASR (Automatic Speech Recognition): The process of converting spoken language into text using machine learning and natural language processing.

ROC (Receiver Operating Characteristic): A curve that shows the tradeoff between the true positive rate and false positive rate across different decision thresholds.

DET (Detection Error Tradeoff): A curve that plots false rejection rate against false acceptance rate, emphasizing performance in error tradeoffs.

REST (Representational State Transfer): A way of defining services or APIs. Stateless, separates client and server concerns, and uses standard HTTP methods (like GET, POST, PUT, DELETE) to interact with resources identified by URIs.

URI (Uniform Resource Identifier): A unique web address used to identify and access a specific resource on the internet.

API (Application Programming Interface): A set of defined rules and protocols that allow different software applications to communicate and interact with each other. It acts as an intermediary, enabling one application to request services or data from another, and receive responses.

Microservice: A set of defined rules and protocols that allows different software applications to communicate and interact with each other. It acts as an intermediary, enabling one application to request services or data from another, and receive responses.

III. References

- [1] OpenAI, “New AI classifier for indicating AI-written text,” Jan. 31, 2023. [Online]. Available: <https://openai.com/blog/new-ai-classifier-for-indicating-ai-written-text>
- [2] S. Gehrmann, H. Strobelt, and A. M. Rush, “GLTR: Statistical Detection and Visualization of Generated Text,” in *Proceedings of ACL 2019 System Demonstrations*, 2019, pp. 111–116.
- [3] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning, and C. Finn, “DetectGPT: Zero-Shot Machine-Generated Text Detection using Probability Curvature,” in *NeurIPS 2023 Datasets and Benchmarks Track*, 2023. [Online]. Available: <https://arxiv.org/abs/2301.11305>
- [4] B. Sheng, X. Wu, K. Tang, Y. Li, and Y. Chen, “Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated Text via Conditional Probability Curvature,” 2024. [Online]. Available: <https://arxiv.org/abs/2310.05130>
- [5] K. Krishna, Y. Song, M. Karpinska, J. Wieting, and M. Iyyer, “Paraphrasing Evades Detectors of AI-Generated Text, but Retrieval Is an Effective Defense,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [6] J. Kirchenbauer, J. Geiping, Y. Wen, et al., “A Watermark for Large Language Models,” 2023/2024. [Online]. Available: <https://arxiv.org/abs/2301.10226>
- [7] X. Wang, J. Yamagishi, M. Todisco, et al., “ASVspoof 2019: A Large-Scale Public Database of Synthesized, Converted and Replayed Speech,” *Computer Speech & Language*, vol. 64, 2020, Art. no. 101114.
- [8] H. Tak, J. Patino, A. Nautsch, et al., “End-to-End Anti-Spoofing with RawNet2,” in *Proc. ICASSP 2021*, 2021, pp. 6369–6373.
- [9] J.-W. Jung, J. Heo, H. Tak, H.-S. Heo, and J.-S. Chung, “AASIST: Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks,” in *Proc. ICASSP 2022*, 2022, pp. 6367–6371.
- [10] J. Yi, J. Tao, X. Tian, et al., “ADD 2023: The Second Audio Deepfake Detection Challenge,” in *DeepFake Audio Detection and Analysis Workshop (DADA) at IJCAI*, 2023.
- [11] P. Howell and J. Kadi-Hanifi, “Comparison of Prosodic Properties Between Read and Spontaneous Speech,” *Speech Communication*, vol. 10, no. 2, 1991, pp. 163–169.

- [12] E. Blaauw, "The Contribution of Prosodic Boundary Markers to the Perception of Spontaneous and Read Speech," *Speech Communication*, vol. 15, no. 1–2, 1994, pp. 131–146.
- [13] J. E. Cole (ed.), "Prosody in Spontaneous and Read Speech—Some Differences," *Journal of the Acoustical Society of America*, vol. 109, no. 5 (Suppl.), 2001.
- [14] E. Shriberg, "Phonetic Consequences of Speech Disfluency," SRI International Technical Report, 2007.
- [15] M. Nakamura, T. Oba, K. Shikano, and T. Kawahara, "Differences Between Acoustic Characteristics of Spontaneous and Read Speech and Their Effects on Speech Recognition," *Computer Speech & Language*, vol. 22, no. 2, 2008, pp. 171–184.
- [16] R. Smith, "AI Is Forcing the Return of the In-Person Job Interview," *The Wall Street Journal*, Aug. 12, 2025. Accessed Sep. 11, 2025.
https://www.wsj.com/lifestyle/careers/ai-job-interview-virtual-in-person-305f9fd0?gaa_at=eafs&gaa_n=ASWzDAjHNCXib6rTmQeBUrx2MUluMuBdEty44rsOp1tGh6af-qMHT3F_67o14aUVaDc%3D&gaa_ts=68c37a60&gaa_sig=EMOEX__fF-mbM9uhHcH_0zANRa6vERIXxwYhgQKcXgppXb4rYh8XL3-ZvOILC1DcJ-LDEaG2_DCBQbrHyD_8yA%3D%3D
- [17] W. Liang, L. Li, M. Baz, et al., "GPT Detectors Are Biased Against Non-Native English Writers," *Patterns*, vol. 4, no. 7, 2023, Art. no. 100779.